

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	02 October 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

The following functional requirements define the core behaviour of the proposed ServiceNow-based Automated Network Request Management solution. They are derived from the project's Service Catalog journey, documented Workflow Studio implementation, approval path, record updates and notification actions. Capabilities not visible in the implementation evidence are treated as planned or future scope.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Request Catalog	Provide a Service Catalog item for employees/users to submit network-related requests. Acceptance: Capture the configured request information and create a ServiceNow Requested Item (RITM) from the submission.
FR-2	Request Data Capture	Capture required request information through catalog variables and mandatory fields. Acceptance: Configured variables are available to the workflow; incomplete required information cannot be submitted.
FR-3	Dynamic Form Behaviour	Use clear form behaviour to show or require fields according to the request context. Acceptance: Conditional visibility and mandatory behaviour work as configured without unnecessary client-side logic.
FR-4	Request Submission & Workflow	Trigger the Network Request Flow Designer process when the catalog request is submitted. Acceptance: The Service Catalog trigger receives the Requested Item Record and starts the configured flow.
FR-5	Catalog Variable Retrieval	Retrieve selected variables from the Network Request catalog item for downstream automation. Acceptance: The Get Catalog Variables action exposes the configured variables as flow outputs.
FR-6	Operational Record Creation	Create a structured operational record from the submitted request values. Acceptance: Configured catalog values are mapped into the operational record and Approval Status is initialized as Requested.
FR-7	Approval Management	Send the created network request record through the configured approval action. Acceptance: An approval request is generated and the configured approver/approval rule can provide an outcome.
FR-8	Approval Decision & Branching	Evaluate the approval outcome and execute the correct workflow path. Acceptance: Approved requests follow the Approved branch; non-approved requests follow the Else path and receive the corresponding record update.
FR-9	Request Notifications	Send lifecycle notifications at configured request stages. Acceptance: The request-created notification is sent after record creation and the configured resolution notification is sent after the approved update.
FR-10	Request Status & Tracking	Maintain an understandable request status across the workflow. Acceptance: The operational record reflects the configured lifecycle state such as Requested, Approved or Rejected.
FR-11	Fulfilment Handoff & Operations	Make the approved request context available for downstream network fulfilment and operational tracking. Acceptance: The approved request contains the mapped request information needed by the responsible operational process; detailed fulfilment automation is a future extension unless configured.
FR-12	Workflow Monitoring & Handoff Documentation	Support Flow Designer execution review, troubleshooting and solution handoff. Acceptance: Administrators can inspect execution details, mappings and condition paths and maintain documentation for recreation in another ServiceNow instance.

Non-functional Requirements:

The non-functional requirements define quality expectations for usability, security, reliability, performance, availability, scalability, maintainability, data integrity, compatibility and auditability. These are application-level expectations; platform-level guarantees depend on the deployed ServiceNow instance.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The requester-facing catalog experience should be simple, readable and guided. Mandatory fields and clear descriptions should reduce incomplete submissions.
NFR-2	Security	Access to network-request and operational data should be governed through ServiceNow roles and ACLs, with administrative configuration restricted to authorized users.
NFR-3	Reliability	For valid submissions, catalog data should move consistently from the Service Catalog through Flow Designer into the configured operational record.
NFR-4	Performance	Normal request submission and workflow processing should avoid unnecessary manual steps and excessive client-side processing; server-side orchestration should remain maintainable.
NFR-5	Availability	The solution is expected to be accessible whenever the deployed ServiceNow instance and required platform services are available; no independent uptime guarantee is asserted by this project.
NFR-6	Scalability	The design should allow additional network request categories, variables, approval rules, fulfilment paths and notification rules without redesigning the complete solution.
NFR-7	Maintainability	Prefer Service Catalog, Flow Designer, UI Policies and ACL configuration. Custom scripting should be limited to necessary behaviour and the workflow should remain understandable to another administrator.
NFR-8	Data Integrity	Values captured in the catalog should remain consistent when mapped to the operational record. Mapping and validation errors should be detectable during testing.
NFR-9	Compatibility	The requester experience should operate in the intended ServiceNow Service Portal/catalog context and use supported platform capabilities for workflow and notifications.
NFR-10	Auditability & Troubleshooting	Administrators should be able to inspect workflow execution, approval/condition paths, stored values and available execution information to diagnose failures.

Future / Out-of-Scope Requirements for the Current Version

- Additional network request categories and specialized catalog items beyond the currently demonstrated Network Request item.
- Multi-level approval chains such as manager approval followed by network-security approval, unless configured in the implementation.
- Automated network-device/API integrations for direct configuration changes.
- Automated fulfilment task creation and closure rules beyond the workflow currently demonstrated.
- Advanced SLA calculation, escalation and performance dashboards.
- External messaging channels such as SMS, WhatsApp or other notification integrations.
- AI-based request classification, priority prediction or anomaly detection.
- Advanced network analytics and operational dashboards.

Requirement Traceability Summary

The critical end-to-end requirement path is: requester enters valid network-request information → Service Catalog submission → Flow Designer variable retrieval → operational record creation with Requested state → approval → approval evaluation → Approved/Rejected update → notification → operational fulfilment and tracking. This path defines the highest-priority testing and troubleshooting scenarios.